

1(a) Direct slide

(b) slide

(c) reduces both bias and variance

Bias: starts with a weak learning. But at each instance more focus is given to misclassified instances adjusting subsequent models to perform better; gradually improving accuracy of overall model.

Variance: diverse set of models, each captures a different regions of the data. Overall, they provide a stable prediction.

Example: any (maybe email spam)

(b) H_0 : low variance, high bias

H_1 : high variance, low bias

we can choose

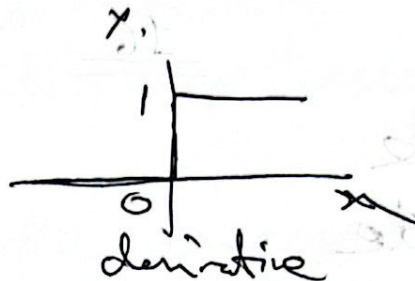
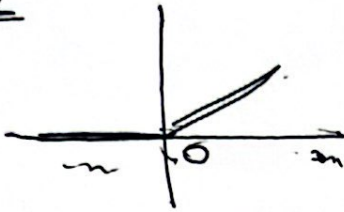
H_0 . less complex

(overfit)

may not completely fit the two given points ~~but~~ and may generalize more.

3(c) slide.

5(a)



Not diff at $x=0$.

We can use $\frac{1}{2}$ at that point. Very rare case.

(b) slide.

(c) $\mu_1 = 3$ $\sigma_1 = 1.22$ $\mu_2 = 2.5$ $\sigma_2 = 1.118$ $\mu_3 = 8.75$ $\sigma_3 = 0.433$ $\mu_4 = 5.25$ $\sigma_4 = 1.479$ $\mu_5 = 2.5$ $\sigma_5 = 1.118$.

BN $\rightarrow \gamma \cdot \frac{x_i - \mu}{\sigma} + \beta$

1st row $\rightarrow [-0.81, -1.34, 0.58, 1.18, -1.34]$

For LN

across features $\rightarrow$

$\mu = 4$

$\sigma = 3.347$

$[-0.597, -0.896, 1.49, 0.896, -0.896]$

6(b) slide

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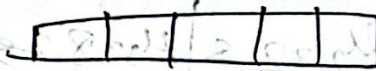
1(a) slide.

(b). slide.

(c)(i) label $\longleftrightarrow$ no label.

(ii) val $\rightarrow$ simply divide in train and validation set.

cross-val $\rightarrow$



train on $(k-1) + 1$ test

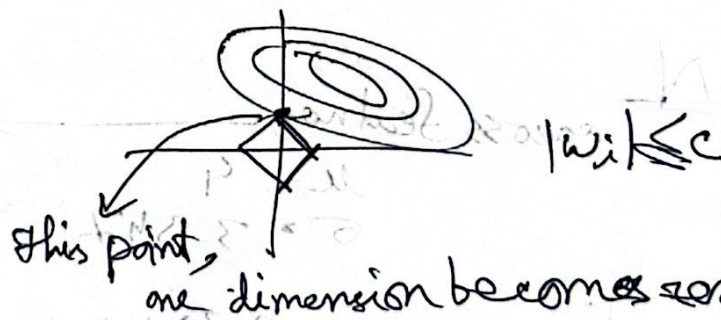
(see slides + lecture)

take avg. of all the estimations.

(iii) slide.

2(a). slide

sparsity.



In general, in geometric interpretation, the constraint regions, contours and region intersect at coordinate axes which makes one or more small values zero.

2(b) slide

(c) MAP $\rightarrow$ adv:
simple and tractable - suitable
for large hypothesis space.

disadv:
can make dangerous predictions,
can go to decisions too early.

3(a) slide

(b) slide

(c) see drive. full solution, (Tanjeem)

5. (a) give yours

(b). $\begin{bmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_k \end{bmatrix}$
substituted

$$\frac{\partial L(Y, \hat{Y})}{\partial \hat{y}_k} = \frac{\partial}{\partial \hat{y}_k} \sum_{j=1}^K y_j \log \hat{y}_j$$

$$= \frac{y_k}{\hat{y}_k}$$

Ans

$$\sum_{i=1}^K \frac{y_k}{\hat{y}_k}$$

$$(c) \frac{\delta f}{\delta w_0} = -\frac{\ln}{\cancel{\log}}(1 + e^{-(w_0 x_0 + w_1 x_1 + b)}) x_0$$

$$= -0.365$$

$$\frac{\delta f}{\delta w_1} = -\ln(1 + e^{-(\quad)}) x_1$$

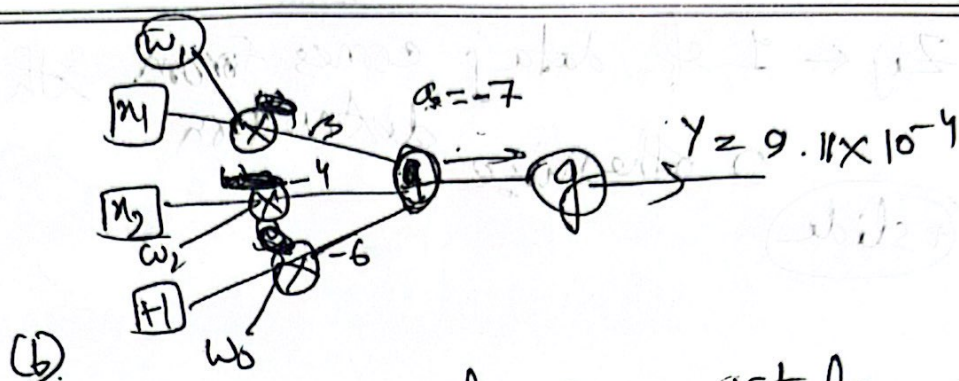
$$= 0.547$$

$$\frac{\delta f}{\delta b} = -\ln(\quad) \cdot 1$$

$$= -0.91$$

8(c) ??

1(a)



(b) Same like Sharif sir 1st class math

2(a)

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) q_t$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) q_t^2$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2}$$

$$\theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{t} + \epsilon} \left(\beta_1 \hat{m}_t + \frac{(1 - \beta_1) q_t}{(1 - \beta_1)} \right)$$

look ahead with $\hat{m}_t$

↑ bias correction

for regular momentum $\rightarrow \theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{t} + \epsilon} \hat{m}_t$

Add NAG
to Adam

4(b) $Z_{ij} \leftarrow 1$ if data j comes from i th Gaussian distribution
0 otherwise
(+ slide)

5(a) max variance
↓
data are spread out on the principal axes
↓
more info captured
↓
less discrepancy with original data
↓
min error.

6(b) slide

6(a) slide

7(a) cost $J \rightarrow$ quantifies difference between predicted and actual values.

problems:

- multiple local minima
- plateaus, saddle point
- vanishing gradient.

7(c) slide.

7(c) ^{on} dataset with limited samples on uneven class distribution.

adv:

1. maximizing the use of data.

2. we can train on all regions of the data ~~and~~ and choose the hyperparameters likewise.
• In ~~the~~ 2-split, some regions may be missing.

3. In 2-split, ~~the~~ performance can vary depending on the random selection of data points.

8(a) $\frac{d}{dn} (\log(n)) = \text{sigmoid}(n)$

importance $\rightarrow$ slide.

'8(c) slide

1.(a) $\Rightarrow$ Stock price prediction
Temperature μ

$$F1score = 2 \times \frac{PV \times TPR}{PV + TPR}$$

(b) same as before

3.(i) ~~NOT same~~

$$\nabla_{\theta} J(\theta) = -\frac{n}{\sqrt{r_t + \epsilon}} p_{m,t-1}$$

$$m_t = \beta m_{t-1} + (1-\beta) \text{ ~~} \nabla_{\theta} J(\theta) \text{ }~~$$

$$r_t = \bar{r}_t + q_t^2$$

$$\theta_{t+1} = \theta_t - \frac{n}{\sqrt{r_t + \epsilon}}$$

4.(c) ~~NOT~~ in syllabus

5.(a) (b) (c) slide.

6.(a) slide

~~$$(b) P(a|h) = \frac{n_1}{2} + n_2 ll + 2n_3 ll + n_4 \left(\frac{1}{2} - 3ll \right)$$~~

$$\frac{\delta P}{\delta ll}$$

$$(b) \quad p(d|h) = \left(\frac{1}{2}\right)^{n_1} \ell \ell^{n_2} (2\ell \ell)^{n_3} \left(\frac{1}{2} - 3\ell \ell\right)^{n_4}$$

$$L(d|h) = n_1 \log\left(\frac{1}{2}\right) + n_2 \log \ell \ell + n_3 \log 2\ell \ell + n_4 \log\left(\frac{1}{2} - 3\ell \ell\right)$$

$$\frac{dL(d|h)}{d\ell} = 0 + \frac{n_2}{\ell \ell} + \frac{n_3}{\ell \ell} + \frac{-3n_4}{\frac{1}{2} - 3\ell \ell} = 0$$

$$\frac{1}{\ell \ell} (n_2 + n_3) = \frac{6n_4}{1 - 6\ell \ell}$$

$$\ell \ell \cdot 6n_4 = (n_2 + n_3) (1 - 6\ell \ell)$$

$$\ell \ell \cdot 6n_4 = n_2 + n_3 - 6(n_2 + n_3)\ell \ell$$

$$\ell \ell = \frac{n_2 + n_3}{6(n_2 + n_3 + n_4)}$$

$$\frac{n_2 + n_3}{6}$$

7(b) slide

8(a)

$$A \rightarrow \text{bone} \rightarrow B(0.4) \approx 0.97$$

$$\text{allu} \rightarrow 0.7 B(0.5714) + 0.3 B(0)$$

$$= 0.689$$

$$\text{gain} = 0.28$$

$$B \rightarrow \text{before info} = B(0.4) = 0.92$$

$$\text{after} = 0.4B(0.75) + 0.6B\left(\frac{11}{6}\right)$$

$$= 0.4 \times 0.811 + 0.6 \times 0.6$$

$$= 0.71$$

$$\text{Crain} \approx 0.74$$

(1) slide.